

V3 empirical study · Instruction

Two models · four underlyings · four volatility regimes · 1.5-year monthly calibration · option-implied Heston fami

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of two models tracks realized stock price S_t most closely, and does the ranking change across companies and the four volatility regimes?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t . That row is bold and shaded green; Ranking ranks 1-2.
- Companion stock-path report. This file keeps Heston and Heston-Merton. Rankings are within this pair; they are not compared here with return-based models.
- The other stock companion is V4_1p5y_monthly_stock_price_return_based.ipynb.

Experimental design (held fixed for every model)

- Models: Heston (no jumps), Heston-Merton (Bates).
- Companies: SPY, AAPL, MSFT, AMZN. Each name is calibrated and simulated separately.
- Regimes: Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- P-measure: keep the lookback drift μ . Do not replace $\mu \rightarrow r$ (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

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Path construction

- One rolling path cloud per ticker $\times$ regime $\times$ model, $n_paths = 10000$, seed 42.
- Clock: daily trading days from the first bar of the regime through the regime end. $\Delta t = 1/252$.
- Start at the observed open S_0 . Refresh parameters at month-ends with data $\leq$ that date.
- At each day, summarize the cloud by p10, p25, p50, p75, p90. The reported path is p50, not the Monte Carlo mean.

Metrics

- Primary: $RMSE\% = 100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. Ranking uses this only.
- $MAE\% = 100 \times \text{mean}(|p50 - S_t|/S_t)$. $Bias\% = 100 \times \text{mean}((p50 - S_t)/S_t)$.
- Tables report RMSE%, MAE%, Bias%, and Ranking only.

Page 3 · Table block · SPY · two models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	26.46	20.99	+20.03	2
Heston-Merton	25.88	20.28	+19.02	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	4.48	3.72	+3.71	2
Heston-Merton	4.25	3.52	+3.52	1

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	9.68	8.49	+8.49	2
Heston-Merton	9.58	8.29	+8.29	1

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	8.41	6.40	-1.44	1
Heston-Merton	8.61	6.67	-1.93	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 4 · Table block · Company summary · two models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green = lowest mean RMSE% vs realized S_t. 1.5-year lookback, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis

2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	36.24	28.22	+19.90	2
Heston-Merton	35.08	27.64	+17.62	1

Table Co-2 Company summary · 2013-2014 Normal

2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	19.03	16.95	+5.60	1
Heston-Merton	19.04	16.93	+4.80	2

Table Co-3 Company summary · 2018-2019 Late-cycle

2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	26.10	23.89	+23.89	2
Heston-Merton	25.26	22.91	+22.90	1

Table Co-4 Company summary · 2019-2020 COVID

2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	18.80	15.40	-14.45	1
Heston-Merton	19.53	15.99	-15.11	2

RMSE%, MAE%, and Bias% are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-simulation of a combined price series. Ranking uses the mean RMSE% only.

Page 5 · Table block · AAPL · two models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	53.57	42.24	+39.16	2
Heston-Merton	51.96	41.02	+36.75	1

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	17.48	14.90	-13.26	1
Heston-Merton	18.37	15.64	-13.93	2

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	30.62	27.11	+27.09	2
Heston-Merton	29.66	25.87	+25.84	1

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	25.43	22.72	-22.70	1
Heston-Merton	26.13	23.30	-23.28	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 6 · Table block · MSFT · two models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	17.91	15.09	+6.02	2
Heston-Merton	17.83	15.04	+5.10	1

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	4.06	3.42	-2.44	1
Heston-Merton	4.53	3.84	-2.95	2

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	12.11	10.39	+10.39	2
Heston-Merton	11.85	9.90	+9.90	1

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	10.50	8.63	-7.46	1
Heston-Merton	11.33	9.28	-8.20	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 7 · Table block · AMZN · two models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	37.25	27.34	+14.53	2
Heston-Merton	35.47	26.87	+11.00	1

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	35.55	32.51	+32.50	2
Heston-Merton	34.22	31.32	+31.29	1

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	35.57	34.18	+34.18	2
Heston-Merton	34.27	32.96	+32.96	1

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
Heston	20.48	14.84	-13.20	1
Heston-Merton	21.14	15.38	-13.86	2

RMSE% is $100 \times \sqrt{\text{mean}((p50 - S_t)/S_t)^2}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Heston-Merton	21.57	19.75	10
2	Heston	21.85	19.19	6

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
Heston	1	2	2	1	6
Heston-Merton	3	2	2	3	10

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
Heston	33.80	15.39	22.00	16.21	21.85
Heston-Merton	32.78	15.34	21.34	16.80	21.57

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
Heston	12.26	9.05	1
Heston-Merton	12.08	9.09	3

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
Heston	31.78	28.03	2
Heston-Merton	31.53	27.89	2

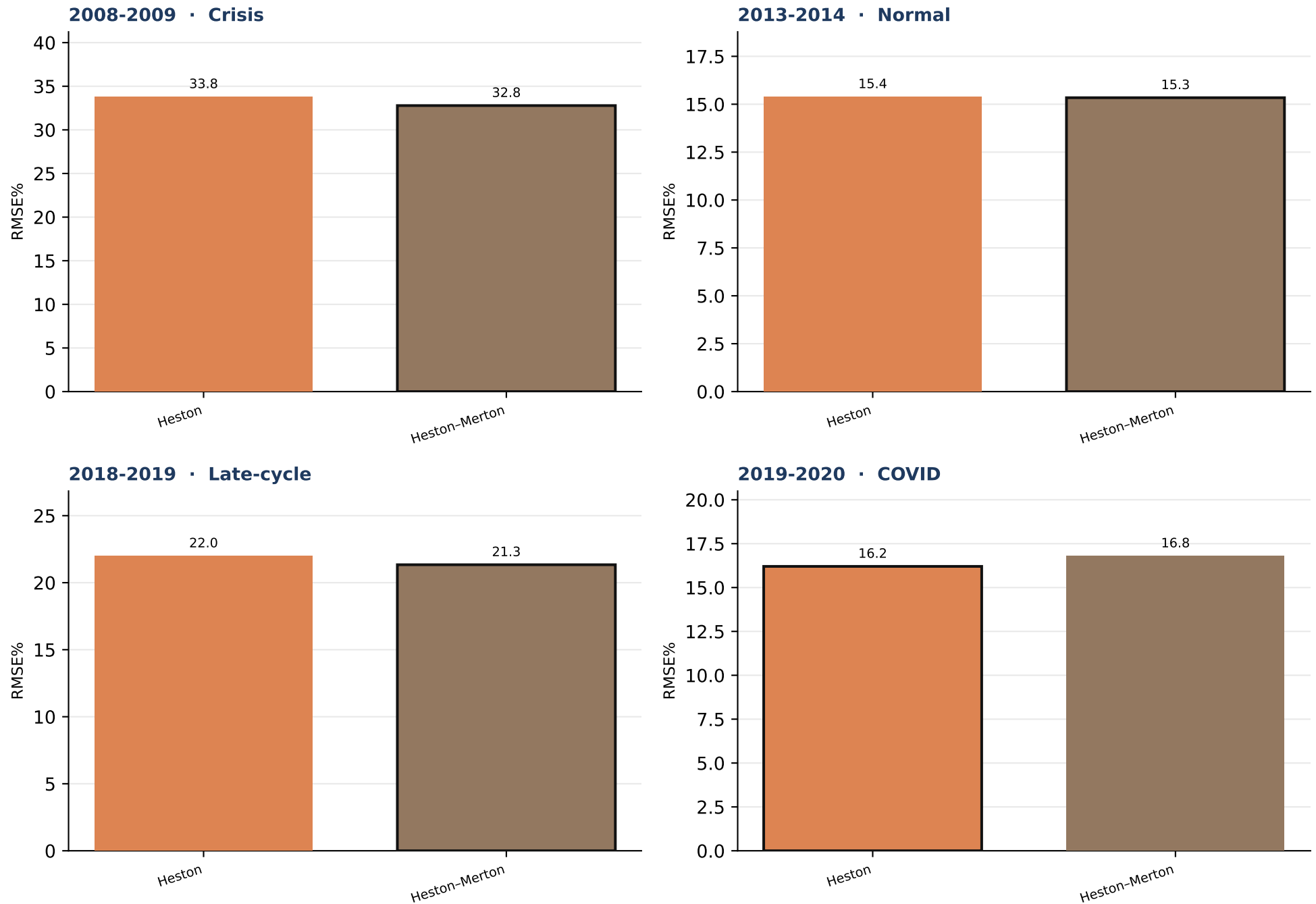
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
Heston	11.14	11.31	2
Heston-Merton	11.39	11.59	2

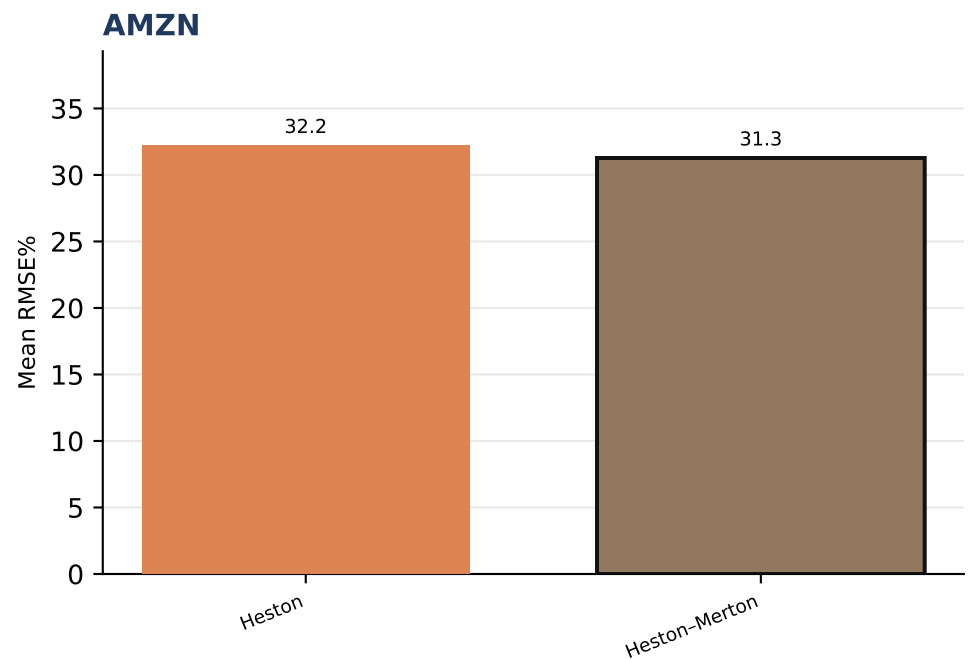
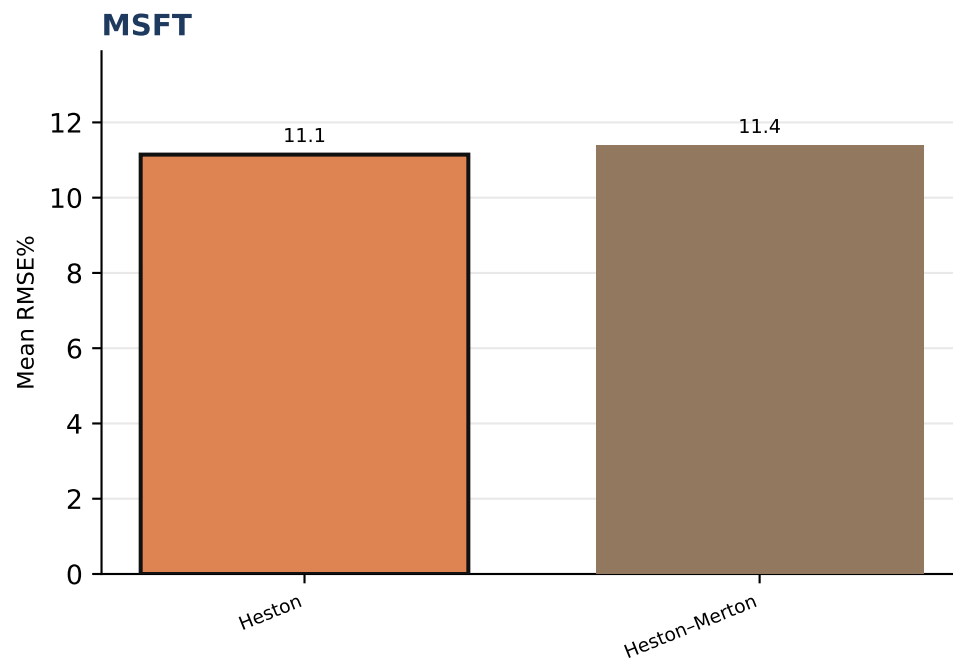
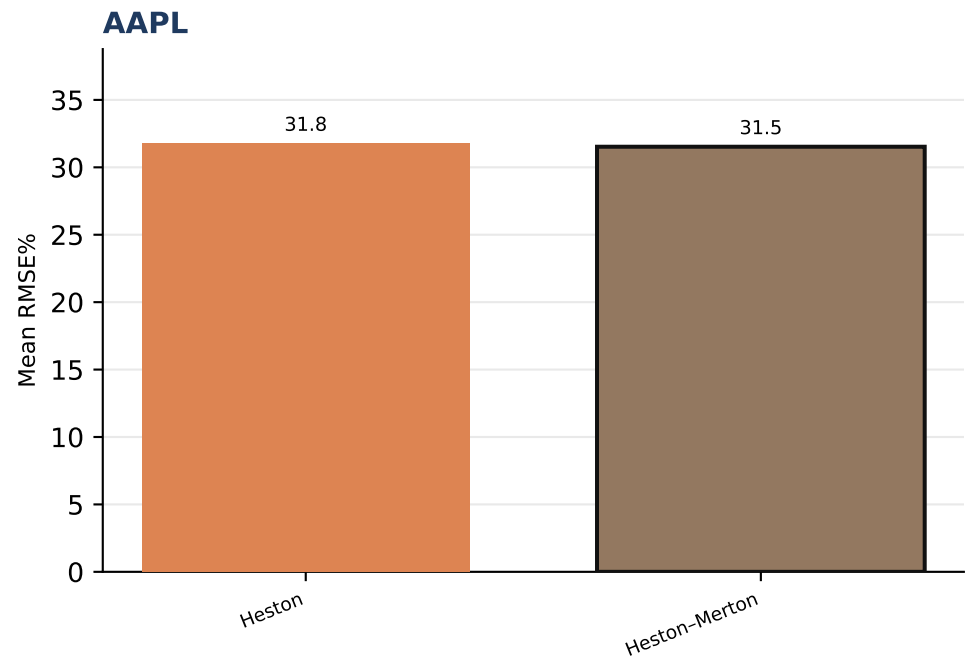
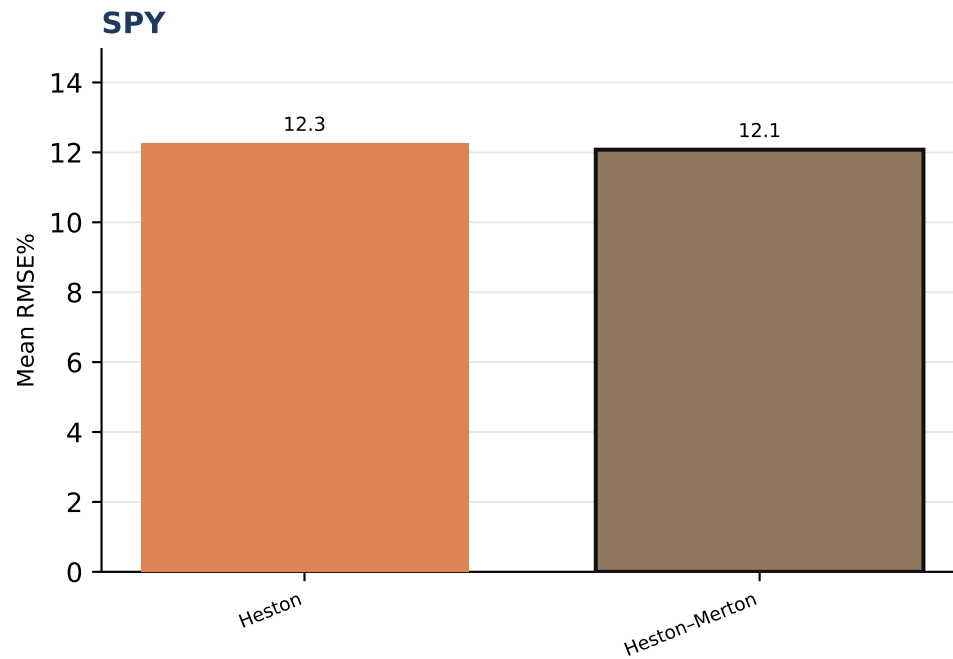
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
Heston	32.21	35.56	1
Heston-Merton	31.27	34.25	3

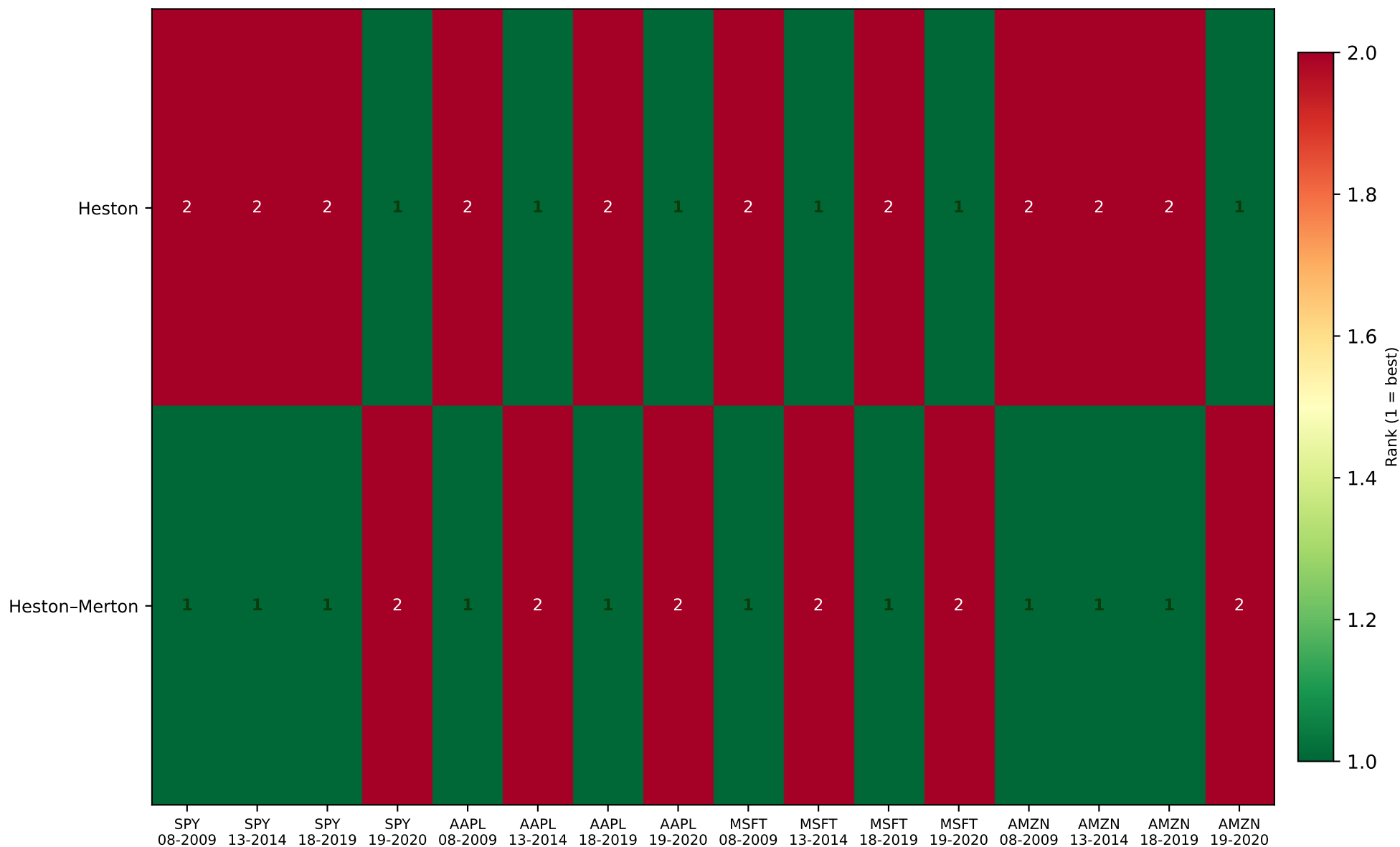
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



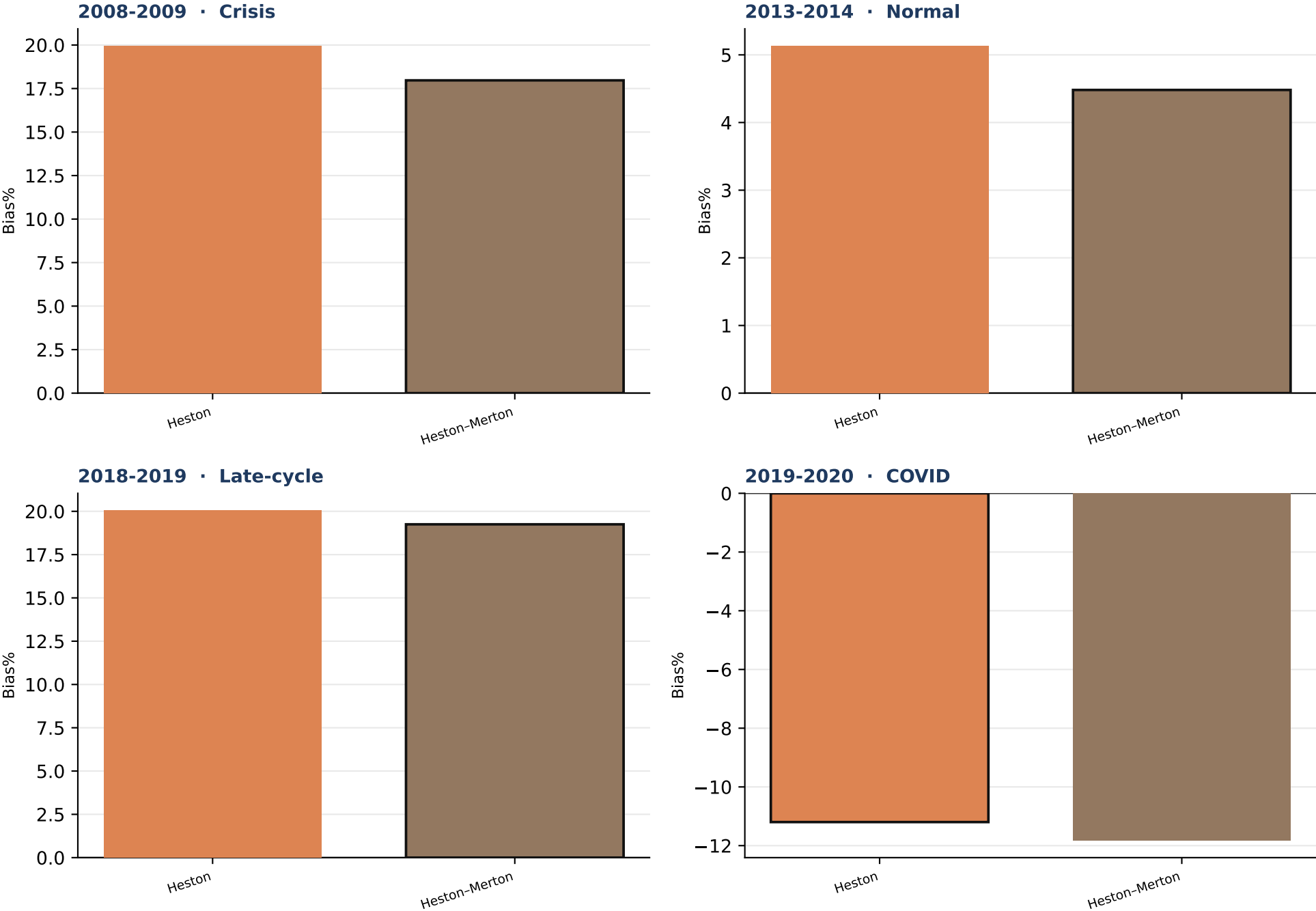
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero Bias% (not the RMSE ranking). Positive = model expensive vs listed quote.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 16 company×regime cells, Heston–Merton is best (mean RMSE% = 21.57; 10 of 16 cells).
- The ranking is not stable. Best-in-cell models: Heston, Heston–Merton.
- These conclusions are conditional on 1.5-year monthly calibration P-measure calibration, 10000 paths, daily steps, and the p50 scoring convention.